

BBTN Olfaction Lecture Handout

Giacomo Glotzer

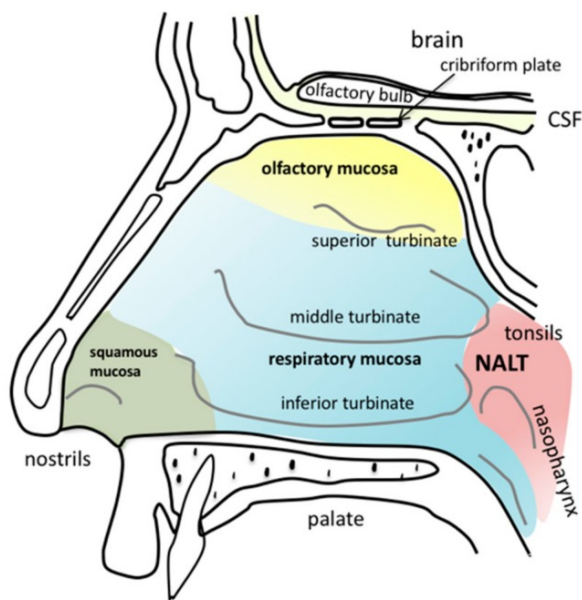
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Goals for today

- Understand the difficulty of approximating odorant perceptual space
- Understand the limitations of an aqueous environment in mediating olfaction
- Gain comfort with Henry's Law

Part 1. Getting to know the nose (35min)

1.1 Geometry of the nose (5min)



Olfactory sensory neurons (OSNs) are bipolar neurons with ciliated dendrites in the olfactory epithelium (OE) and axons projecting to olfactory bulb.

Steps required to smell something:

1. Volatilization

1. odorant enters the headspace above its source; depends on temperature, vapor pressure, and source concentration.

2. Deposition in the OE

1. depends on airflow patterns shaped by the turbinates, and on solubility-dependent deposition along the mucosa.

3. Receptor activation

1. odorant-binding proteins (OBPs) ferry hydrophobic odorants through aqueous mucus to the receptor; depends on receptor affinity (K_d).

4. Neural signaling

1. OSNs fire and activate downstream synaptic partners in olfactory bulb glomeruli.

To activate an OR, an odorant must cross from the air phase into an aqueous environment. Insects pick up molecules directly by contact on externalized organs, but vertebrates require odorants to first volatilize and then dissolve into nasal mucus.

1.2 How much mucus sits in our olfactory epithelium? (5min)

Human OE surface area $\approx 5 \times 10^2 \text{ mm}^2 = 5 \times 10^{-4} \text{ m}^2$

Mucus layer $\approx 20 \text{ } \mu\text{m} = 2 \times 10^{-5} \text{ m}$ thick (treated as water):

$$V_{\text{mucus}} = 5 \times 10^{-4} \text{ m}^2 \times 2 \times 10^{-5} \text{ m} = 10^{-8} \text{ m}^3 \times \left(\frac{10^2 \text{ cm}}{\text{m}} \right)^3 \times \frac{1 \text{ mL}}{\text{cm}^3} \times \frac{10^3 \text{ } \mu\text{L}}{\text{mL}} = 10 \text{ } \mu\text{L}$$

Ten microliters; the same volume everyone has pipetted.

1.3 How long would it take our mucus to evaporate? (10min)

Model evaporative flux as:

$$\dot{m} = K \times A \times \Delta P$$

$$K = 2 \times 10^{-2} \text{ mg} \cdot \text{m}^{-2} \cdot \text{s}^{-1} \cdot \text{Pa}^{-1}$$

K is the mass transfer coefficient for water evaporating into still air.

Dimensional analysis

Remember dimensions of pressure:

- $P = \frac{F}{A} = \frac{\text{M} \cdot \text{L} / \text{T}^2}{\text{L}^2} = \frac{\text{M}}{\text{L} \cdot \text{T}^2} = \text{M L}^{-1} \text{T}^{-2}$
- $\text{Pa} = \frac{\text{N}}{\text{m}^2} = \frac{\text{kg} \cdot \text{m} \cdot \text{s}^{-2}}{\text{m}^2} = \text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2} = \text{M L}^{-1} \text{T}^{-2}$

Plugging in for the units (mg, m, s, Pa) in K:

$$K = \text{mg} \cdot \text{m}^{-2} \cdot \text{s}^{-1} \cdot \text{Pa}^{-1} = M \cdot \frac{1}{\text{L}^2} \cdot \frac{1}{\text{T}} \cdot \frac{\text{LT}^2}{M} = \frac{\text{T}}{\text{L}}$$

(Note that mass units are carried through K implicitly by M/M)

$$\dot{m} = \frac{T}{L} \times L^2 \times \frac{M}{L \cdot T^2} = \frac{M}{T}$$

So evaporative flux is giving us a mass over time.

At 25°C, $P_{\text{sat}} \approx 3 \times 10^3 \text{ Pa}$. This is the saturation vapor pressure of water, or the partial pressure of water vapor when the air is in equilibrium with a liquid water surface (i.e., 100% RH).

At 20% relative humidity (RH):

$$\Delta P = P_{\text{sat}}(1 - \text{RH}) = 3 \times 10^3 \text{ Pa} \times 0.8 \approx 2 \times 10^3 \text{ Pa}$$

Rate of evaporation:

$$\dot{m} = \frac{2 \times 10^{-2} \text{ mg}}{\text{m}^2 \text{ s Pa}} \times (5 \times 10^{-4} \text{ m}^2) \times (2 \times 10^3 \text{ Pa}) \approx 2 \times 10^{-2} \text{ mg/s}$$

Time to evaporate all 10μL (10mg):

$$t = \frac{m}{\dot{m}} = \frac{10 \text{ mg}}{2 \times 10^{-2} \text{ mg/s}} = 500 \text{ s} \approx 10 \text{ min}$$

So an unprotected mucus film would evaporate in roughly 10 minutes under dry ambient conditions.

How we resist desiccation:

1. **Bowman's glands** continuously secrete serous fluid to replenish the mucus layer.
2. **Mucins** are heavily glycosylated proteins that are hygroscopic and resist evaporative water loss far better than a simple aqueous film.
3. **Humidification**. By the time inhaled air reaches the OE it has already been warmed and humidified by the turbinates and respiratory mucosa, substantially reducing ΔP .
4. **Regeneration if all else fails**. OSNs are among the only neurons in the mammalian nervous system that naturally regenerate throughout life, so even if the epithelium dries out, recovery is possible.

Camel Noses

Camel nasal turbinates are highly scrolled, providing $\sim 1000 \text{ cm}^2$ of surface area. During inhalation, moist nasal membranes humidify and cool incoming air. During exhalation, the cooled surfaces reabsorb water vapor, so exhaled air leaves at $\sim 75\text{--}80\%$ RH rather than the 100% it had in the lungs, saving roughly 60–70% of respiratory water loss.

1.4 What is the maximum molecular weight of an odorant? (15min)

Ants communicate using **cuticular hydrocarbons (CHCs)** — straight-chain alkanes and alkenes ranging from C20–C40 (~350–520 Da). These molecules encode nestmate identity, caste, and reproductive status with extraordinary specificity. Yet humans cannot smell them at all. Why not? Can we derive the upper MW limit of human odorants from first principles?

We restrict ourselves to **straight-chain alkanes** because their physical properties scale cleanly with carbon number.

Longer alkanes have lower vapor pressure (meaning their partial pressure will be lower) and partition more strongly out of water (meaning they prefer air to water) because they are even more hydrophobic.

$$C \uparrow: MW \uparrow: \text{hydrophobicity} \uparrow: P_{vap} \downarrow: \text{Solubility} \downarrow$$

Saturated vapor expressed as molar concentration: $[\text{air}]^{\text{max}} = \frac{P_{vap}}{RT}$

$$\text{Saturated mucus: } [\text{mucus}]^{\text{max}} = \frac{[\text{air}]^{\text{max}}}{K_{aw}} = \frac{P_{vap}/RT}{P_{vap}/(S \cdot RT)} = S$$

$$\text{Where } K_{aw} = \frac{K_H}{RT} = \frac{[\text{odorant}]_{\text{air}}}{[\text{odorant}]_{\text{water}}} = \frac{P_{vap}}{S \cdot RT}$$

$$[C]_{\text{mucus}}^{\text{max}} = S(n)$$

The most you can ever deliver to the mucus from a pure source is just the **aqueous solubility** of the alkane. This makes physical sense: a puddle of pure alkane saturates the air at P_{vap} , and air saturated at P_{vap} is in equilibrium with water saturated at S .

How fast does alkane solubility drop with chain length, and where does it fall below the detection threshold?

Let's take the most sensitive human ORs, with $K_D \approx 100$ nM, and require 10% occupancy:

$$[C]^{\text{threshold}} \approx 0.1 \times K_D = 10 \text{ nM} = 10^{-8} \text{ M}$$

Each added CH₂ group costs a roughly fixed free energy of transfer into water, so solubility drops by a roughly constant factor per carbon, about **10× every ~2 carbons**

Alkane	Carbons	S (aqueous, M)
Octane	C8	$\sim f \times 10^{-6}$
Decane	C10	$\sim f \times 10^{-7}$
Dodecane	C12	$\sim f \times 10^{-8}$
Tetradecane	C14	$\sim f \times 10^{-9}$

We need $S(n) \geq 10^{-8}$ M. From the table:

- **C12:** $S \approx 2 \times 10^{-8}$ M $\rightarrow$ just above threshold
- **C14:** $S \approx 1 \times 10^{-9}$ M $\rightarrow$ below threshold

The limit sits between C12 and C14.

Tetradecane (C14, MW = 198 Da) sits at the edge of detectability in human psychophysics experiments (Doleman & Lewis 1998, PNAS).

This means that beyond C14, alkanes are practically unsmellable by the airborne route.

How do ants sense >C30 CHCs?

1. Contact chemosensation
2. Waxy cuticle
3. OBPs shuttle CHC through the lymph
4. Receptor tuning

The chemical language available to a species is set by its peripheral sensory architecture not just its receptor repertoire.

Part 2. Dogs vs Humans (25min)

2.1 Can a human or a dog smell "better"? (5min)

Sensitivity depends on:

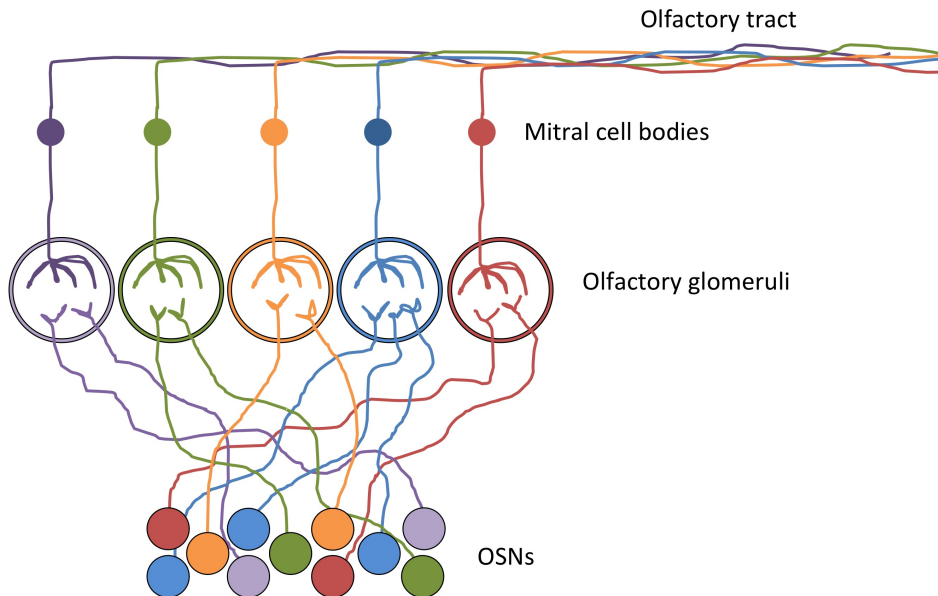
- proximity to source
- sniff volume
- OE surface area
- number of OSNs
- number of ORs per OSN (OSNs typically carry multiple cilia)
- receptor K_d

Discrimination depends on:

- number of distinct OR types
 - number of glomeruli
 - downstream processing
-

2.2 The canonical model (5min)

Canonical model: 1 receptor type : 1 neuron : 1 glomerulus.



2.3 How many more OSNs does a dog have? (5min)

Note: $N_{\text{OSN}} \gg N_{\text{OR}}$ across all species.

Humans ^[1]

$$N_{\text{OSN}} = \underbrace{2 \times 10^4 \text{ mm}^{-2}}_{\text{density } (\rho)} \times \underbrace{5 \times 10^2 \text{ mm}^2}_{\text{OE area}} = 10^7$$

Roughly 10^7 (10 million) OSNs.

Dogs ^[2]

Breeds like Labradors, German Shepherds, and hounds have elongated snouts with $\sim 10^2 \text{ cm}^2$ of OE:

$$N_{\text{OSN}} = 2 \times 10^4 \text{ mm}^{-2} \times 10^4 \text{ mm}^2 = 2 \times 10^8$$

Roughly 2×10^8 (200 million) OSNs, or an order of magnitude more than humans.

Dogs can apparently detect odorants in the PPT range for some odorants. ^[3] How is this possible?

2.4 How many more ORs does a dog have? (5min)

OSN number sets a ceiling on sensitivity but does not determine how many odors can be discriminated. That depends on the number of distinct OR types and the overlap of their tuning curves.

Species	Functional ORs	Pseudogenes
Human	~400	~400
Dog	~800	—
Mouse	~1200	—
Fly	~60	—
Clonal raider ant	~500	—

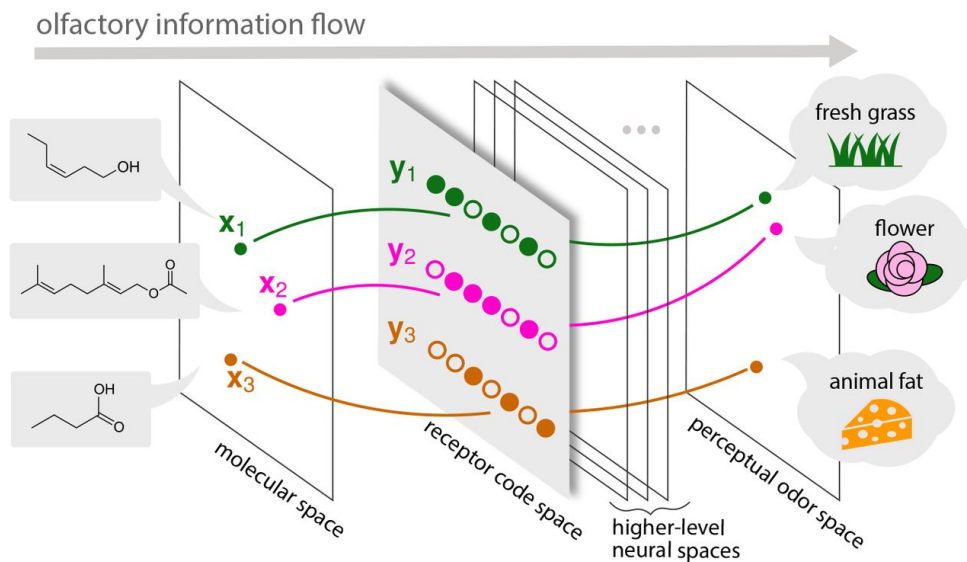
This means the number of OSNs per OR in dogs is 10x higher than that of humans.

Why do humans have so many OR pseudogenes? One hypothesis: the transition to trichromatic color vision in the primate lineage relaxed selection on olfaction, [ref.](#) Old World monkeys (trichromats like chimps, gorillas, macaques) show higher rates of OR pseudogenization than New World monkeys (dichromats like tamarins, marmosets), except for the howler monkey (NWM that independently evolved trichromacy & shows elevated pseudogene fractions). But the paper has issues (correction) and two authors retracted their names.

Insect aside: OR repertoire size in insects correlates with whether a lineage is winged. Across at least three of four independent losses of winged flight in hymenopteran females, there are corresponding expansions of the OR repertoire. Flightless insects may rely more heavily on chemosensation to navigate their environment.

2.5 How many more odorants can a dog discriminate? (5min)

Figure: molecular space → receptor code space → higher-level neural spaces → perceptual odor space



Molecular vs receptor vs perceptual space

- Receptor code space is the bottleneck. We can actually measure and quantify the number of receptors, their sensitivities, and the number of cells that express them.
- Higher-level neural spaces are tough to access and can expand dimensionality enormously through combinatorics.
- Perceptual odor space is also difficult to quantify. Approaches include discrimination tasks (animals rewarded for correctly telling odors apart) and human psychophysics.

Floor: number of ORs (channels)

Ceiling: $2^{N_{\text{glomeruli}}} = 2^{N_{\text{OR}}}$ which if you have 100 ORs $\approx 10^{30}$

- You can have narrowly tuned receptors which are specialists and easy to interpret or broadly tuned receptors with overlapping tuning curves
- Even a modest receptor repertoire supports discrimination of a vast number of compounds through combinatorial coding where each odorant activates a particular pattern of ORs and therefore a particular pattern of active glomeruli.
- Bushdid, Vosshall and Magnasco claimed humans can discriminate >1 trillion odorant mixtures but this was later refuted by Markus Meister in a rebuttal paper that showed that their math was an artifact of inferring a ball-filling geometry.
- Open question: are mixtures treated as distinct odor "objects," or decomposed into their components?

2.6 Could a receptor be sensitive to a mixture of two odors? (5min)

[primacy coding paper](#)

Some ORs may be tuned to mixtures rather than to individual compounds.

The **primacy set** is the small group of highest-affinity ORs that activate first at low concentration and remain active at higher ones. The hypothesis is that odor identity is encoded by which ORs belong to this set, not by the full activation pattern, making the code concentration-invariant.

From the paper (Eq. 1–2), the activation of receptor r by a mixture of D odorants follows mass-action binding:

$$\frac{f_r}{1 - f_r} = \sum_{i=1}^D \frac{c_i}{K_{d,ri}}$$

where:

- f_r is the fractional occupancy of receptor r — the fraction of receptor molecules bound to odorant. We assume response is proportional to occupancy.
- c_i is the concentration of odorant i
- $K_{d,ri}$ is the dissociation constant of receptor r for odorant i — lower means tighter binding

Each term $c_i/K_{d,ri}$ asks: how saturated is this receptor by odorant i ? The terms add across all odorants in the mixture.

Half-maximal activation ($f_r = 1/2$) is reached when $\sum_i c_i/K_{d,ri} = 1$. A receptor is "in the primacy set" if it crosses this threshold at the lowest total mixture concentration. The choice of $1/2$ as the threshold is a convenience — the primacy ranking holds for any threshold, as long as it is the same for all receptors.

Imagine three receptors:

- **OR1** binds pinene only, $K_d = 100 \mu\text{M}$
- **OR2** binds limonene only, $K_d = 100 \mu\text{M}$
- **OR3** binds both, $K_d = 150 \mu\text{M}$ for each (weaker, but broad)

For pure pinene, OR1 wins ($K_d = 100 \mu\text{M}$) over OR3 ($K_d = 150 \mu\text{M}$ for pinene alone). Same for pure limonene: OR2 wins. OR3 is not in the primacy set for either pure odorant.

50/50 mixture at total concentration C (so $c_{\text{pin}} = c_{\text{lim}} = C/2$).

OR3 receives drive from both components; OR1 and OR x2 each see only one:

$$\left(\sum_i \frac{c_i}{K_{d,ri}} \right)_{\text{OR3}} = \underbrace{\frac{C/2}{150 \mu\text{M}}}_{\text{geraniol}} + \underbrace{\frac{C/2}{150 \mu\text{M}}}_{\text{limonene}} = \frac{C}{150 \mu\text{M}}$$

$$\left(\sum_i \frac{c_i}{K_{d,ri}} \right)_{\text{OR1}} = \frac{C/2}{100 \mu\text{M}} + 0 = \frac{C}{200 \mu\text{M}}$$

Setting each equal to 1 gives the total concentration needed for half-occupancy:

When $\frac{f_r}{1-f_r} = 1 \rightarrow f_r = 0.5$

Receptor	C needed for $f_r = 1/2$
OR3	150 μM
OR1, OR2	200 μM

OR3 reaches half-occupancy first, at a lower total mixture concentration than the specialists. It enters the primacy set for this mixture, even though it is a weaker binder to each individual component. A broadly-tuned OR with moderate affinity for multiple components can outcompete a specialist because the mixture provides additive drive.

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1. Estimated from OE area $\approx 5 \text{ cm}^2$ and OSN density $\approx 2 \times 10^4 \text{ mm}^{-2}$. See: <https://www.ncbi.nlm.nih.gov/books/NBK556051/> ↩
 2. Estimated from OE area $\approx 100 \text{ cm}^2$ in long-snouted breeds and similar OSN density. See: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8375464/> ↩
 3. Detection threshold data: <https://www.sciencedirect.com/science/article/abs/pii/S0168159105002194> ↩